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02443 - Stochastic Simulation Project Report 1 - Group 40

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Part 1

The model follows women after they had their breast tumor removed. The cancer may reappear close to the removed tumor. This is called local recurrence, and the woman enters state 2. The cancer may also reappear distant from where it was operated. This is called distant metastasis. Both things may also occur. Death can occur from any state. In this model, once the death state has been entered, it can never be left again. This means the simulation should be terminated once this state has been reached.

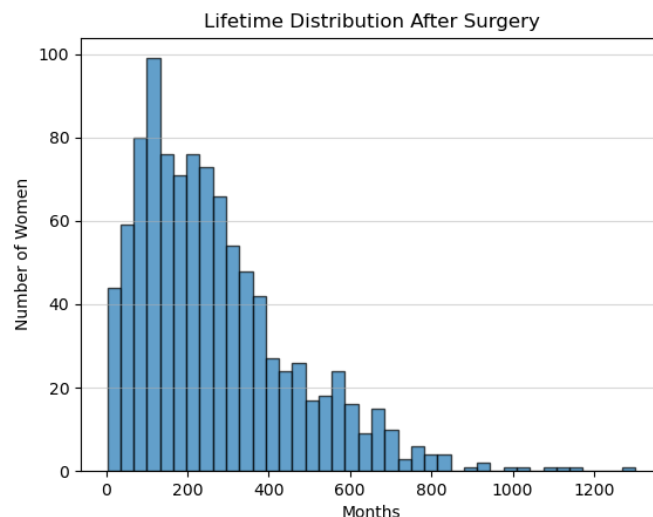
Task 1

Assume that one time step equals one month. Simulate 1000 women, all starting in state 1, until death. Summarize the lifetime distribution of the women, after surgery, for example using a histogram. In what proportion of women does the cancer eventually reappear, locally?

Solution:

We simulated the markov chain transitions and got the following results

Proportion of women with local recurrence: 0.6980



Task 2

In your simulations, what is the distribution over the states at $t = 120$? Does this correspond to what we expect? Answer the question using an appropriate statistical test.

Solution:

Validation at $t = 120$ months:

Expected Counts: [359.03 158.96 166.09 67.74 248.19]

Observed Counts: [352. 181. 169. 70. 228.]

Chi-square Statistic: 4.9633

P-value: 0.2911

Result: Fail to reject the null hypothesis. The simulation matches the analytical distribution.

Task 3

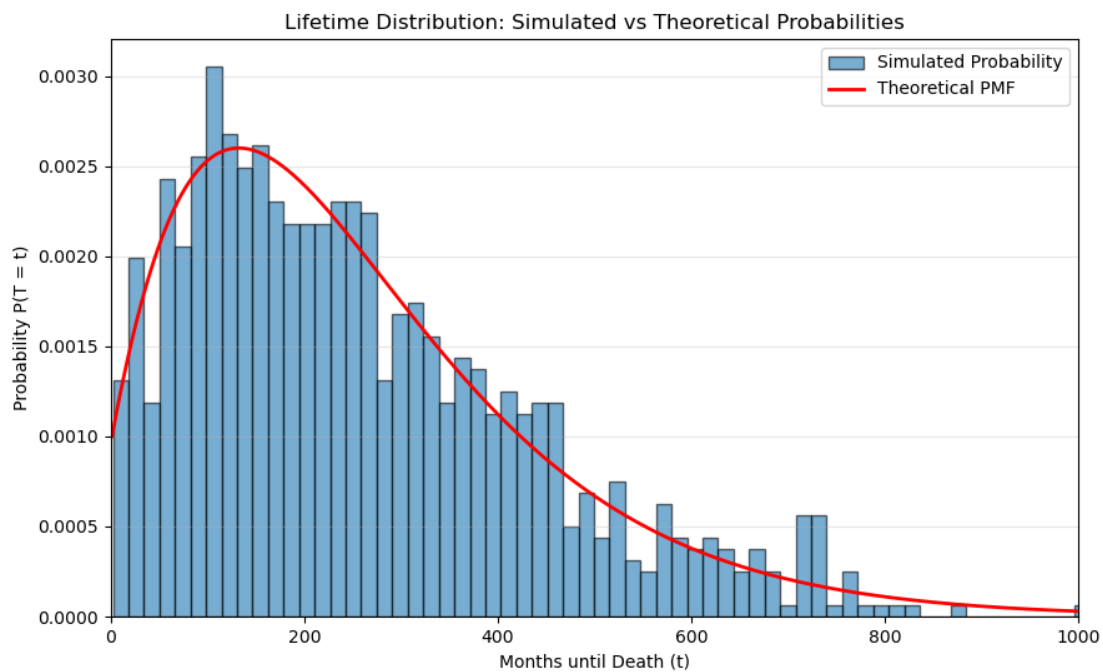
Does your simulated lifetimes follow this distribution?

Solution:

From the plot and the values below it seems that our simulation pretty much follows the expected distribution.

Theoretical Mean Lifetime: 262.37 months

Simulated Mean Lifetime: 263.16 months



Task 4

Estimate the expected lifetime, after surgery, of a woman who survives the first 12 months following surgery, but whose breast cancer has also reappeared within the first 12 months, either locally or distant.

Solution:

Using the rejection sampling method as suggested in the project task, we got the following result.

Expected lifetime (conditional): 164.49 months Found 1000 matches after 11850 attempts.

Task 5

What fraction of women die within the first 350 months? Answer this by simulating 200 women, and record the fraction. Do this 100 times. Use control variates to reduce the variance. How large a reduction in variance do you see, using control variates, as opposed to the crude Monte Carlo estimator?

Solution:

By using the control variate reduction method with the mean value of the sample being the control variable, we got the following results

Crude Estimator Mean: 0.7418

Crude Estimator Variance: 0.000917

CV Estimator Mean: 0.7392

CV Estimator Variance: 0.000355

Variance Reduction: 61.33%

Task 6

For the report, consider the following questions: What assumptions underlie the discrete time Markov chain model? Are those assumptions realistic? How may we relax some of these assumptions, possibly at the cost of increased model complexity?

Solution:

The first assumption that the Markov chain model uses is that the process is memory-less and that the probability distribution of the next time step doesn't depend on the past values of the process, which is true in case of exponential distribution and binomial distribution. In our breast cancer model too we assume that the process is memoryless and that's why we can use the Markov chain model here.

There's also an assumption we made here is that we're using just discrete states and are moving from one to another in a fixed one month timeline which isn't very realistic but is easier to simulate and study the evolutions.

Though these assumptions seem restrictive, in real life a lot of processes are almost memory-less and that's why Markov chain models are so useful in real world simulations, though if the distribution depends somewhat on the past states and we use Markov chains we introduce inherent errors in the simulation.

The benefit of using Markov chain model is that we don't need to keep track of the past states to be able to predict the next state, which makes the simulation pretty straightforward, on the other hand if we were to use an exact model which needs previous states too to predict the next state - the state space grows exponentially with time and become computationally impossible for longer simulations.

Part 2

Task 7

Simulate 1000 women, all starting in state 1, until death. Summarize the lifetime distribution after surgery, for example in a histogram. Report the mean, along with a confidence interval, and the standard deviation, also with a confidence interval. In what proportion of women has the cancer reappeared distantly after 30.5 months?

Solution:

Using the simulation method as described in the task, we got the following results

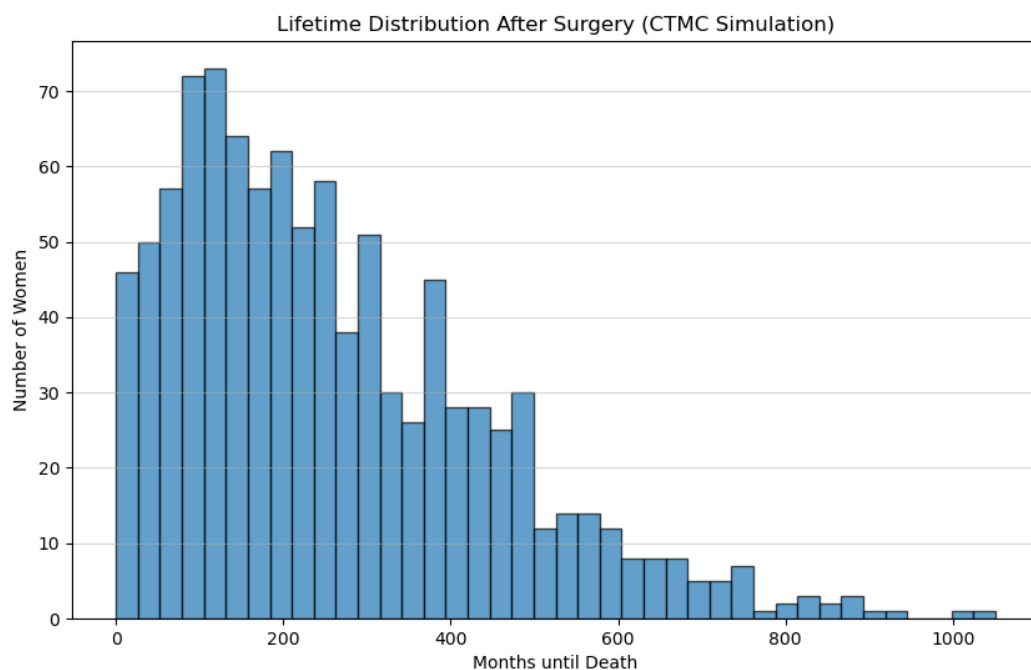
Mean Lifetime: 259.77 months

95% CI for Mean: [248.15, 271.38]

Standard Deviation: 187.40 months

95% CI for Std Dev: [179.53, 195.99]

Proportion with distant recurrence after 30.5 months: 0.6130



Task 8

Compare the empirical lifetime distribution function, from your simulations, to the theoretical, using an appropriate statistical test.

Solution:

Using the χ^2 test we got the following results

Chi-Square Goodness-of-Fit Test Results:

Chi-Square Statistic: 8.5092

P-value: 0.4838

Kolmogorov-Smirnov Test Results:

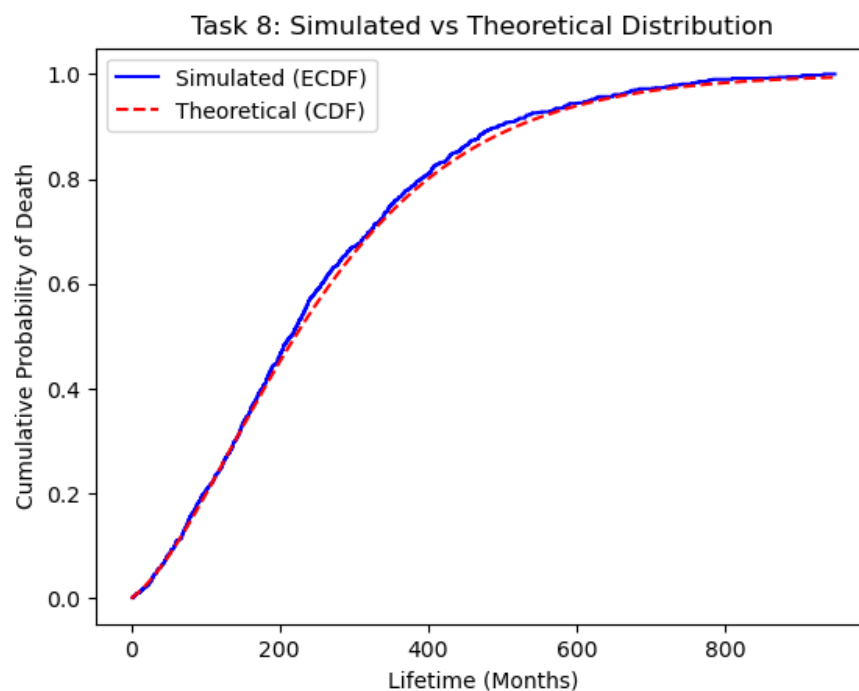
Test Statistic (Max Gap between curves): 0.0275

P-value: 0.4295

The simulation flawlessly matches the phase-type distribution theory!

Conclusion: Fail to reject the null hypothesis.

The binned simulated lifetimes match the expected theoretical frequencies.

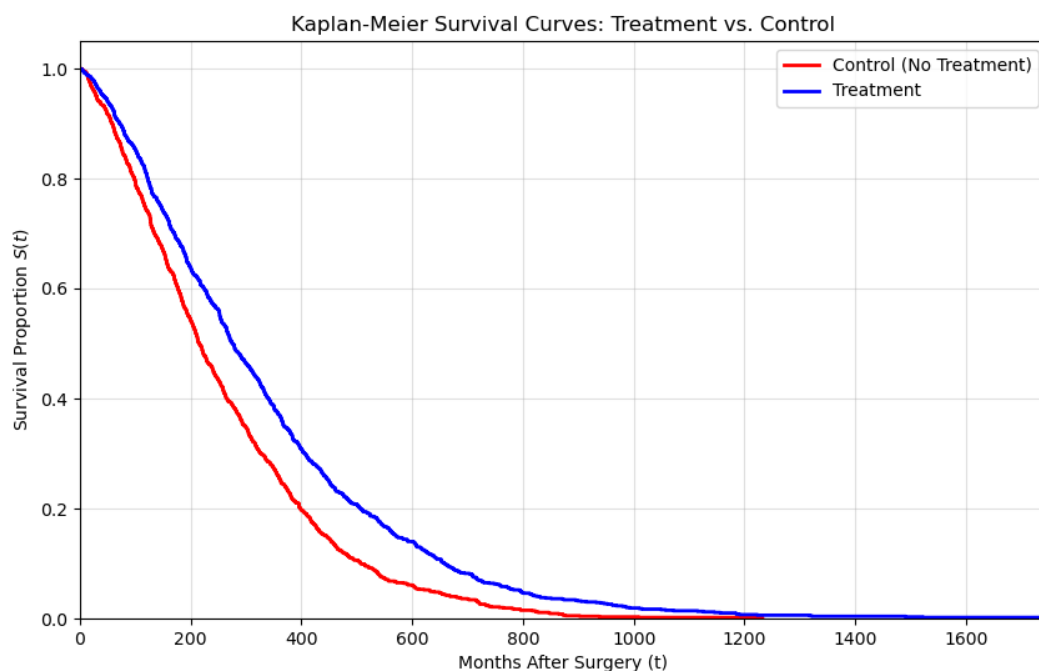


Task 9

Simulate 1000 women who have received this treatment. Plot the KaplanMeier estimate of the survival function. In the same figure, plot the KaplanMeier estimate of the survival function for women, who have not received this treatment. Does the treatment appear to have an effect?

Solution:

From the plot below we can see that the survival proportion of women who're getting the treatment is always higher than the survival proportion of the women who're not on the treatment. This implies that on average a woman on treatment will have higher probability to survive than someone who's not on the treatment. The maximum effect of the treatment is between 200 to 800 months where the survival proportion is much higher.



Task 11

For the report, consider the following questions: What assumptions have been eliminated, by going from the discrete to the continuous time model? What have been added? How could the model be extended, such that the sojourn times are Erlang distributed?

Solution:

We have now removed the assumptions that the transitions are discrete and fixed intervals, rather we're doing continuous time state transitions now which is more realistic.

Even now we're working with memoryless transitions though, the time you spend in each state is an exponential distribution to make the Markov chain model work.

In the discrete model we assumed that transitions can only happen at strict intervals. The continuous model eliminates this restriction, allowing transitions to happen at any exact, real-valued instant

In a discrete step (e.g., one month for the exercises), multiple events could theoretically happen, but the DTMC forces you to summarize them into a single probability. By moving to continuous time, we no longer have to assume that events are aggregated into artificial chunks of time. We can model the exact sequence of events.

To extend the model for Erlang distributed sojourn times we can change the time spent in a state to follow an Erlang distribution and not an exponential distribution which does not closely most medical problems ie where the longer you are sick, the more likely your state is to change

We can use Phase-Type distributions to do the above. To make your Markov model use an Erlang distribution we split the original state into k artificial sub-states (or "phases"). These exponential blocks or phases create a more realistic, bell-shaped waiting time curve.

Part 3

Task 12

Solution:

We sampled the continuous distribution like before and recorded all the state changes along with their time of transition. Then to sample for these discrete values we looked at the state at that point according to the last moved state index.

First 10 simulated time series $Y(i)$:

$Y(1)$: [1, 1, 1, 1, 1, 1, 5]

$Y(2)$: [1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 5]

$Y(3)$: [1, 3, 3, 5]

$Y(4)$: [1, 1, 2, 2, 5]

$Y(5)$: [1, 1, 3, 3, 5]

$Y(6)$: [1, 1, 1, 1, 1, 4, 4, 4, 5]

$Y(7)$: [1, 1, 1, 4, 4, 4, 4, 4, 4, 5]

$Y(8)$: [1, 3, 3, 3, 3, 5]

$Y(9)$: [1, 1, 2, 3, 3, 5]

$Y(10)$: [1, 1, 1, 5]

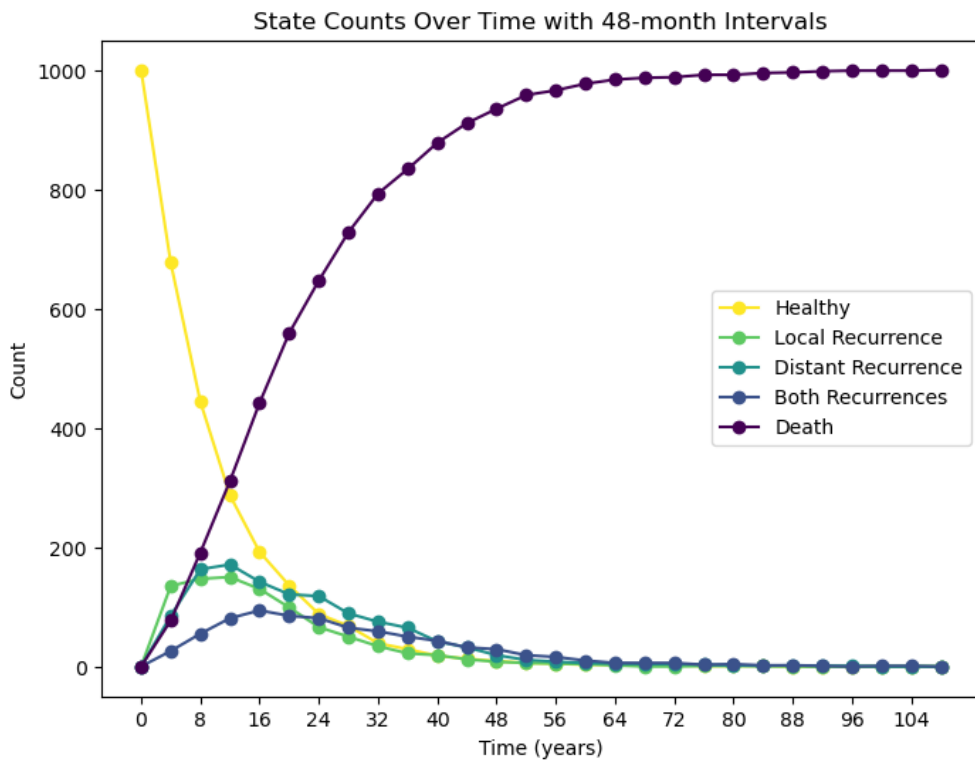


Figure 1: Empirical state distribution visualized

Task 13

Implement the following algorithm to estimate the q_{ij} 's: Select $Q(0)$ as some initial guess. In the k 'th iteration, do the following

- For all of the time series, simulate a possible complete trajectory, taking the observations into account, using $Q = Q(k)$
- Summarize the trajectories
- Find Q^{k+1} using eq 2

Solution:

We implemented the above steps using different values of initial-Q and iterations.

If we start with a guess matrix which has similar structure as the original Q-matrix i.e. 0 below diagonals and 20 max iterations, we get the following result.

Initial Q-Matrix:

$$\begin{bmatrix} -0.04 & 0.01 & 0.01 & 0.01 & 0.01 \\ 0 & -0.03 & 0.01 & 0.01 & 0.01 \\ 0 & 0 & -0.02 & 0.01 & 0.01 \\ 0 & 0 & 0 & -0.01 & 0.01 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Estimated Q Matrix:

$$\begin{bmatrix} -0.0084 & 0.0044 & 0.0026 & 0.0004 & 0.001 \\ 0 & -0.0123 & 0.047 & 0.0034 & 0.0042 \\ 0 & 0 & -0.0078 & 0.0033 & 0.0045 \\ 0 & 0 & 0 & -0.0094 & 0.0094 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Max diff with the true Q Matrix:

0.0016785774925875182

Now if we use a more generic matrix and keep the same max iteration value, we get

Initial Q-matrix:

$$\begin{bmatrix} -0.04 & 0.01 & 0.01 & 0.01 & 0.01 \\ 0.01 & -0.012 & 0.01 & 0.09 & 0.01 \\ 0.01 & 0.08 & -0.11 & 0.01 & 0.01 \\ 0.02 & 0.01 & 0 & -0.34 & 0.3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Estimated Q Matrix:

$$\begin{bmatrix} -0.0087 & 0.0051 & 0.0025 & 0.0004 & 0.007 \\ 0 & -0.053 & 0.0092 & 0.0424 & 0.0017 \\ 0 & 0.0069 & -0.00121 & 0.0034 & 0.0018 \\ 0 & 0.0238 & 0.033 & -0.0427 & 0.0156 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Max diff with the true Q Matrix:
0.039336771774566535

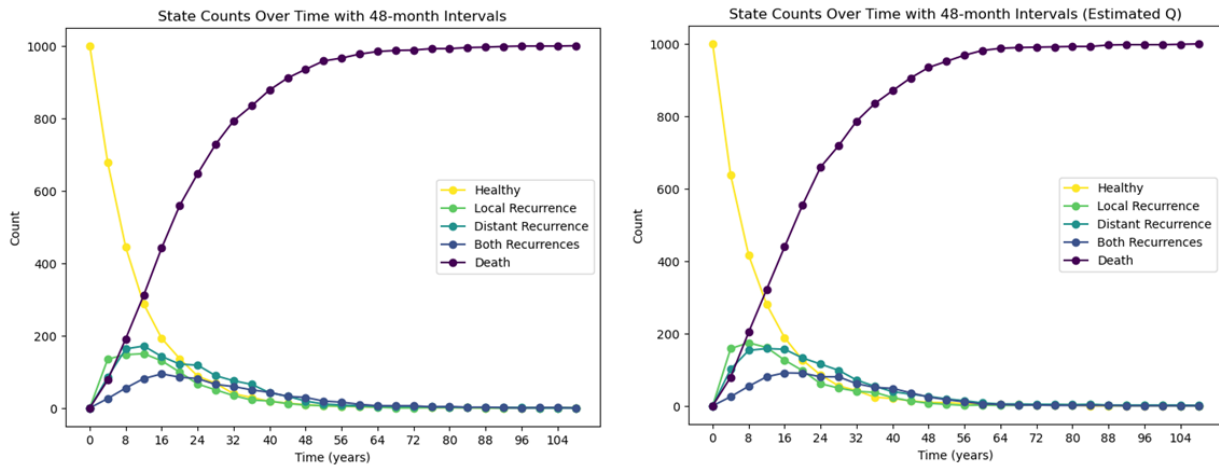
One thing we can notice here is that even though the max difference value is not so large, the actual difference is a lot between the true matrix, the value of transition between state 4 to 1, 2 should be very close to 0 but it's not. This implies that our metric to compare the two matrices is not correct. One idea could be to compare the sum of individual differences instead of maximum difference of values.

Now we run the same initial matrix with 100 iterations and got the following estimation.

$$\begin{bmatrix} -0.0092 & 0.0049 & 0.0031 & 0.000 & 0.0012 \\ 0 & -0.015 & 0.0063 & 0.0087 & 0.00 \\ 0 & 0.0001 & -0.0087 & 0.0026 & 0.006 \\ 0 & 0.0002 & 0.0 & -0.012 & 0.0118 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

We can see that now the values which are supposed to be 0 are much smaller than the previous results. Thus increasing the max iterations to 10x will give us comparable results as when we guesses a matrix with similar structure. It proves that making a good initial guess can give us the same kind of results but will save us a HUGE amount of computational cost.

We used the estimated values to simulate new samples, below you can see the empirical state distributions coming from the original parameters compared to ones coming from the estimated parameters:

Figure 2: Empirical state distribution from estimated Q

We can see that the system behaviour is largely unchanged which means that our estimation models the system that the original observations came from well.